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Abstract

The emergence of plant rust has proven to be a serious threat to agricultural yields. The identification of plant diseases is traditionally done through visual inspection that often lead to delays and significant amount of loss by the time diagnosis is made. On the other hand, advanced deep learning models perform well with accuracy but their large size of datasets and computational complexities make them unsuitable for deployment which limits their real time applications. This study focuses on a method to identify rust using both ground and aerial photographs of plants. It follows a sequence of steps, including pre-processing, extracting Hue, Saturation, and Value (HSV) color features, utilizing Gray-Level Co-occurrence Matrix (GLCM) for extracting texture features, combining these extracted features, and classifying the healthy and rust-affected crops. Machine learning based Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbor (KNN) models are used to achieve higher performance, resulting in greater accuracy in identifying rust pathogens. The performance of our model exceeds existing approaches and is also the highest among feature-based methods, making it the most potential method for rust detection.

Keywords (separated by '-')

Leaf diseases - Color features - Rust Pathogens - Texture feature - Machine Learning

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Integrating Aerial and Ground Level Image Data for Leaf and Stem Rust Detection Using Texture and Color Features

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Abstract. The emergence of plant rust has proven to be a serious threat to agricultural yields. The identification of plant diseases is traditionally done through visual inspection that often lead to delays and significant amount of loss by the time diagnosis is made. On the other hand, advanced deep learning models perform well with accuracy but their large size of datasets and computational complexities make them unsuitable for deployment which limits their real time applications. This study focuses on a method to identify rust using both ground and aerial photographs of plants. It follows a sequence of steps, including preprocessing, extracting Hue, Saturation, and Value (HSV) color features, utilizing Gray-Level Co-occurrence Matrix (GLCM) for extracting texture features, combining these extracted features, and classifying the healthy and rust-affected crops. Machine learning based Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbor (KNN) models are used to achieve higher performance, resulting in greater accuracy in identifying rust pathogens. The performance of our model exceeds existing approaches and is also the highest among feature-based methods, making it the most potential method for rust detection.

Keywords: Leaf diseases · Color features · Rust Pathogens · Texture feature · Machine Learning

1 Introduction

Our society depends on agriculture for their sustainability. In the present situation, where agriculture and plantations are by default an important part of providing food security, the emergence of plant diseases, particularly leaf and stem rust, presents a significant threat to crop productivity. These diseases can severely affect the yield and quality of food crops, leading to economic losses for farmers and disrupting the global food supply. Early and accurate detection

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of such diseases is crucial for implementing timely interventions, reducing the spread of infections, and minimizing crop loss. As the demand for food continues to rise with the growing population, effective disease management strategies are essential for sustainable agricultural practices.

Inconsistency and lag in the recognition of plant diseases cause a reduction in the yielding of quality and good quantity. Traditional methods of disease detection often rely on visual inspections and laboratory testing, which can be time-consuming and inefficient. In contrast, advancements in technology particularly the machine vision possible to fully automate the tasks of visual identification, and the recognition methods can be employed for the successful identification of plant diseases [1,4,13]. The Food and Agriculture Organization (FAO) of the United Nations revealed that total number of starving people has been increasing since 2015 [5]. Due to the plant diseases in the range of 20 to 40% of annual productivity lost globally. In this Study, the assessment was done on different diseases for estimating the root cause of loss. Early and accurate identification of plant diseases is an important factor for ensuring maximum productions and better life for farmers [7].

This paper aims to leverage the technological advancements by developing a robust method for detecting leaf and stem rust using both ground-level and aerial image data. Aerial images have an unique advantage to detect disease symptoms that might not be visible from the ground and is also useful in monitoring large agricultural areas. They are helpful in assessing the crop health more accurately, identifying patterns of diseases and taking timely measures before significant damage occurs. Thus, using dual perspective of the images, a detailed understanding of crop health is acknowledged along with the scope of improving accuracy. Furthermore, using HSV color features and texture analysis helps accurately identify rusted areas, making it easier to distinguish between healthy and diseased plants. By combining both methods, this study highlights the need for better rust detection and aims to support more preventive farming practices [8]. This solution is an innovative approach that secures crop health and guarantees stable food supply. We have proposed a feature-based approach which attempts to overcome the limitations of previous work in terms of accuracy, which is an important factor [1].

Various works in plant disease detection have achieved major success with deep learning models due to their powerful feature extraction capabilities. However, because of high computational requirements, these methods often lack feasibility for mobile applications or real-time field analysis, especially where resources are scarce. Our study undertakes a feature-based approach, combining HSV color features and GLCM texture analysis, which offers a balance of accuracy and efficiency suitable for practical applications. This approach aims to provide a solution that not only improves the accuracy compared to advanced models but also enhances scalability for field use.

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1.1 Objective

While working with different types of datasets for disease detection on integrating all possible features may not be feasible to contribute accurate predictions. This proposed work focuses on developing a system for detecting leaf and stem rust in food crops by leveraging both ground-level and aerial images by extracting critical features such as color and texture features. It can provide efficient solutions for farmers to maintain better crop health.

- **Developing a Robust System.** To detect of leaf and stem rust in food crops using both ground-level and aerial images.
- **Feature Extraction.** To extract relevant features from the images, focusing on HSV color and texture features, to differentiate between healthy and rust affected plants.
- **Machine Learning Implementation.** To train the model and evaluate multiple machine learning models, including Random Forest (RF), Support Vector Machine (SVM), and KNN, to predict the presence of diseases based on the extracted features.
- **Enhance Detection Accuracy.** To utilize the dual perspective of aerial and ground-level images to improve the model's accuracy and efficiency of disease detection, allowing for large-scale monitoring of crop health.
- **Provide a Scalable Solution.** To offer a scalable tool for farmers to productively manage crop health and mitigate the impact of rust diseases on agricultural productivity.

2 Literature Survey

The detection of plant diseases is crucial for nurturing agricultural productivity and ensuring food security. Recent advancements in various technological fields like computer vision, machine learning, and remote sensing have enabled researchers to develop innovative methods for identifying diseases in crops. This section reviews relevant literature that acknowledges the approaches taken, highlights the methodologies, and presents the results of previous studies. Various studies have explored different techniques for detecting specific plant diseases. A two-stage model was developed for extracting color features to identify plant disease. They used SVM for classification, achieving high accuracy, and designed the system for mobile devices with low computational needs [1].

The necessity to overcome challenges like dataset scarcity for rare diseases and real-time image quality was pointed out [2]. Assessment on impact of food security by climate change, economic shock etc. was provided [5]. A hybrid method was proposed for detecting multiple plant diseases by extracting features. Their study faced limitations in dataset diversity, impacting model performance in varied agricultural contexts [6]. Uav-captured images and advanced deep learning models are utilized to detect Cercospora leaf spot(CLS) in sugar beet. The work achieved high accuracy but faced problems due to environmental

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conditions [7]. A method is developed using computer vision and machine learning for automated plant leaf disease detection, focusing on feature extraction and classification [8]. A model was developed for detecting tea leaf blight (TLB) using transfer learning and data augmentation techniques. Their model outperformed conventional methods, demonstrating improved precision and recall. However, its focus on a specific disease limits its generalizability to other diseases or crops [14].

Deep learning is also popular for detecting leaf disease [9]. In this paper, authors were detecting different diseases from mango plant leaves [12]. Some researchers have extracted information from the Medicinal Plant Leaf [10,11]. The recent work in plant disease detection heavily depends on deep learning techniques, particularly CNNs, which have achieved notable success in accurately identifying diseases across various plant species. some Studies reported high classification accuracy for disease detection, relying on large datasets and high computational resources which limits their feasibility for real-time applications [9,12]. In contrast, our feature-based methods which integrate HSV color features and GLCM texture analysis, demonstrate robust performance with lower computational requirements. This study addresses the practical limitations of deep learning models, focusing on a lightweight approach suitable for mobile or handheld device deployment, thus enhancing the accessibility and applicability of rust disease detection tools.

3 Methodology

3.1 Data Preparation

The dataset used, titled *CGIAR Computer Vision for Crop Disease* [3], was accessed from Kaggle. It is pre-divided into 876 training images across three categories - healthy wheat, leaf rust and stem rust and 610 testing images (see Table 1). Some sample images are shown in Fig. 1. This is an in-field collected dataset by the Center of International Maize and Wheat Improvement. In addition, some other data is sourced from publicly available open source images. The goal is to identify the images having wheat condition in the rust form which appears perfectly.

Table 1. Dataset Information

Slno.	Category	Type	Count
1	Diseased(Wheat)	Leaf rust	358
2	Diseased(Wheat)	Stem rust	376
3	Healthy(Wheat)	Healthy	142

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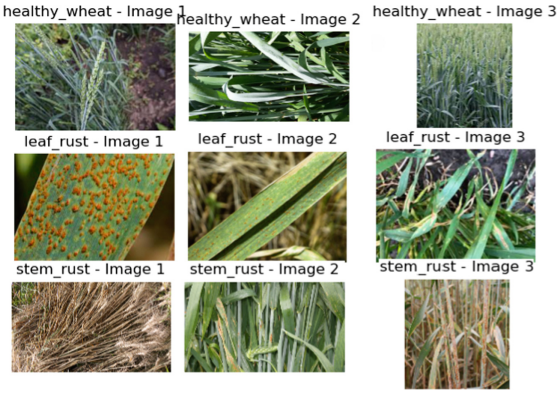

Fig. 1. Images from the dataset

3.2 Data Pre-processing

Images are resized and normalized for the uniform dimension as 224×224 pixels to ensure consistency during training. Brightness adjustments is also applied to enhance the model robustness (see Fig. 2). This annotation help to increase the diversity of the dataset which generalize the model's ability. Red Channel contains the red color intensity values. Green Channel contains the green color intensity values. Blue Channel contains the blue color intensity values.

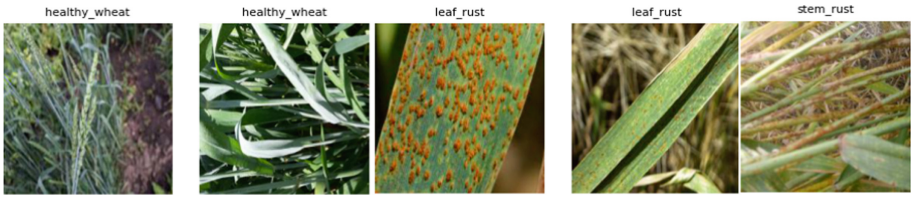

Fig. 2. Resized and Preprocessed data

3.3 Feature Extraction

Color Features Features are extracted using the HSV color space to capture hue, saturation, and value information, which helps differentiate between diseased and healthy areas based on color variations.

Texture Features. Utilizing GLCM to capture texture variations like contrast, homogeneity and dissimilarity associated with rust diseases.

Combining Features. Extracted features were combined the color and texture features into a single feature vector for each image, providing a comprehensive representation of each leaf's characteristics.

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3.4 Feature Combination.

The HSV features are combined with texture features to create a comprehensive feature set. Using PCA to reduce the dimensionality of the feature set to visualize it in 2D or 3D. This minimizes the dimensions of the combined feature set, to easily visualize how the training and testing datasets are distributed in a lower-dimensional space. **Output.**

Combined Train Features Shape: (638, 771)
Combined Test Features Shape: (435, 771)

3.5 Model Training and Evaluation

A classifier is trained with the extracted features. Once this model is trained, the performance is evaluated from the different performance metrics along with the confusion matrix also. RF, SVM and KNN models are considered for this training and evaluation.

Model Training. We utilized three above mentioned machine learning models.

- **SVM.** It is effective for disease classification.
- **RF.** It is robust and well-suited to handle high-dimensional data.
- **KNN.** A strong baseline model for image classification tasks.

In this study, several machine learning classifiers have been chosen as their performance is justifiable for feature-based image classification tasks. Unlike deep learning models, these classifiers utilized the extracted features to achieve decent accuracy while maintaining low computational requirements. By combining HSV and GLCM features, the classifiers in our model achieve results comparable to CNN-based approaches for rust detection. This selection balances accuracy with efficiency, addressing key limitations in scalability and resource dependency that are present in recent advanced deep learning models.

Model Evaluation
For reviewing the performance of our trained model, we have evaluated them on metrics like accuracy, confusion matrix, precision, recall, F1-score for better classification between healthy and rust plants.

- **Accuracy.** Basic measure of correct predictions.
- **Confusion Matrix.** Shows how well your model distinguishes between different diseases.
- **Precision.** Focus on how many predicted positives were correct.
- **Recall.** Focus on catching as many actual positives as possible.
- **F1-score.** Balance between precision and recall.

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4 Feature Extraction Methods

4.1 Color Features (HSV)

For color analysis, images were converted to the HSV color space to capture variations in hue, saturation, and value (see Fig. 3). These features helped to differentiate healthy leaves from diseased ones based on color changes associated with rust infections as shown in the Table 2.

Hue. Represents the color type (e.g., red, green, blue).

Saturation. Indicates the intensity of the color.

Value (Brightness). Reflects the brightness of the color.
The HSV color space is often preferred in image processing because it is nearly the same way humans observe color, making it easier to analyze and manipulate color information effectively.

Table 2. HSV values for each disease type

Disease Type	Hue	Saturation	Value
Healthy Wheat	100.05	53.53	104.03
Leaf Rust	45.31	152.30	124.63
Stem Rust	22.08	97.99	130.60

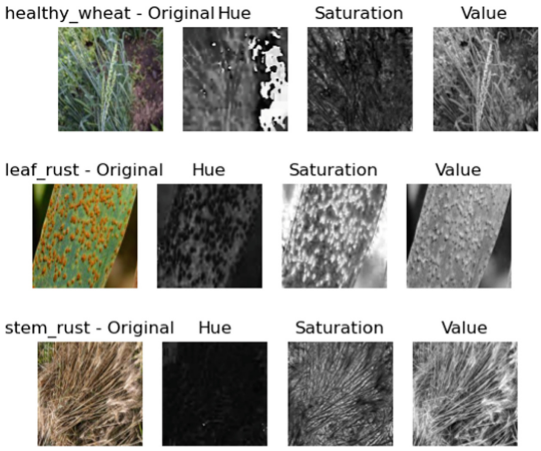

Fig. 3. HSV images along with the original image

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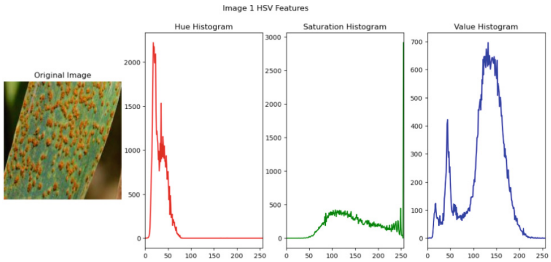
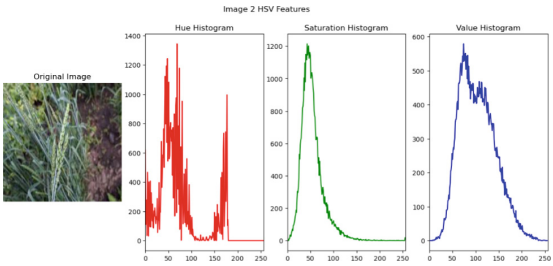
In the Fig. 4, the graphs provide information into the color characteristics of the images in the dataset, highlighting the prevalence of certain hue, saturation, and value ranges. The graph plot defines in what portion the various channels red,blue and green are present in the images .The peak values define the respective color channel to be the highest in the images.



(a) HSV Histogram for diseased image



(b) HSV Histogram for healthy image

Fig. 4. Comparison of HSV Histograms for diseased and healthy images.

4.2 Texture Features (GLCM)

Three texture features are extracted using GLCM to capture surface variations - contrast, homogeneity and dissimilarity. This approach quantified patterns such as smoothness or roughness that often indicate disease presence.

Contrast. It measures the intensity difference between the lighter and darker areas in an image. In the context of image processing, it quantifies how distinct the pixels of different intensities are from each other.

Mathematical Representation. It can be calculated using the difference between the minimum and maximum pixel values in a region or the standard deviation of pixel intensities.

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Homogeneity. It measures the proximity of the distribution of elements in a matrix (like a grayscale image). It indicates how uniform or similar the pixel in intensities are within a region.

Mathematical Representation. It can be calculated using the formula where p(i,j)p(i,j)p(i,j) is the joint probability of intensity levels iii and jjj:

$$\text{Homogeneity} = \sum_{i,j} \frac{p(i,j)}{1 + |i - j|} \tag{1}$$

Dissimilarity. The differences between the pixel intensity values in an image. It quantifies how different the intensity values are from one another.

Mathematical Representation. One common approach to compute dissimilarity is using the following formula where p(i,j)p(i,j)p(i,j) is again the joint probability of intensity levels iii and jjj.

$$\text{Dissimilarity} = \sum_{i,j} p(i,j).|i - j| \tag{2}$$

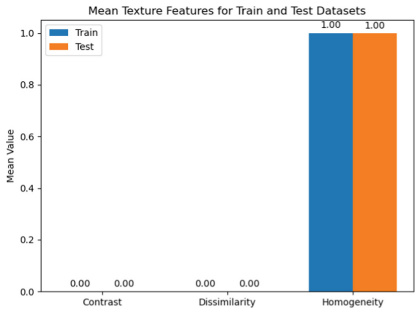

Fig. 5. Graph for texture features

Output.

Train Texture Features Shape: (638, 3)
Test Texture Features Shape: (435, 3)

Figure 5 shows the mean values of texture features (contrast, dissimilarity, and homogeneity) for the training and testing datasets. The values are very low for contrast and dissimilarity, and very high for homogeneity. This suggests that the images in both the training and testing datasets are very similar, with relatively uniform texture and minimal variation. The fact that the mean values are almost identical in the training and testing sets indicates that the two datasets are very well balanced in terms of their texture features, which is a desirable outcome for training a machine learning model.

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5 System Architecture

The System Architecture outlines a procedure for wheat leaf disease detection as shown in Fig. 6. Starting with input data, images undergo image pre-processing to ensure consistency. After that, color and texture features are extracted to capture disease-specific traits. The model training phase then builds a classifier to distinguish between healthy leaves, leaf rust, and stem rust. The prediction and evaluation assesses model performance using metrics like accuracy, precision etc. are shown in the Table 3. This process ensures an accurate and practical system for disease detection.

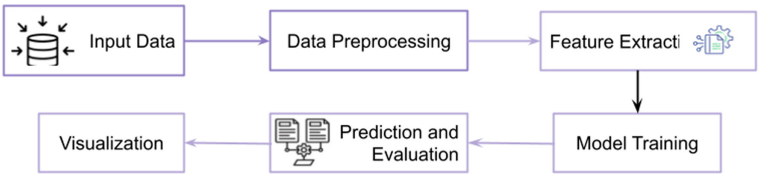

Fig. 6. System Architecture

6 Results

The model is successfully trained and evaluated and RF provides the best accuracy compare to other models (see Table 4).

Table 3. Quantitative evaluation for Random Forest Model

Class	Precision	Recall	F1-score	Support
0	1.00	0.99	1.00	28934
1	0.99	0.99	0.99	12481
Accuracy	0.99			
Macro avg	0.99	0.99	0.99	41415
Weighted avg	0.99	0.99	0.99	41415

Output:

Training Accuracy: 0.99543378
Shape of Test Features: (610, 150528)
Test Predictions: (435, 3)

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Table 4. Evaluation results of the models

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9954	1	1	1
SVM	0.64	0.65	0.48	0.55
KNN	0.6986	0.75	0.58	0.65

Thus, the RF model has the best accuracy among the other models. It is saved and used for correctly predicting healthy and diseased plants. We have provided the correctly predicted results showing our model's good performance that has reaches to distinguish and classify between the healthy and diseased plant.

(a) leaf rust (b) healthy (c) stem rust

Fig. 7. Overview of Images

6.1 Performance Analysis and Comparative Evaluation

In this study, we achieve high accuracy in plant disease detection by combining aerial and ground-level image data and utilizing color and texture features. Table 4 shows an analysis of our model's performance metrics (accuracy, precision, recall, F1 score) that could be compared with models from recent literature. Our feature-based approach with RF achieved an accuracy of 99%, outperforming certain CNN-based models [1] under similar conditions while using fewer computational resources. Although CNNs excel in high-complexity feature extraction, their dependence on large datasets and computational requirements limits their practical application in real-time field scenarios [14]. Our dual perspective provides a balanced approach with high accuracy and lower processing requirements [7], making it more suitable for mobile or low-resource environments.

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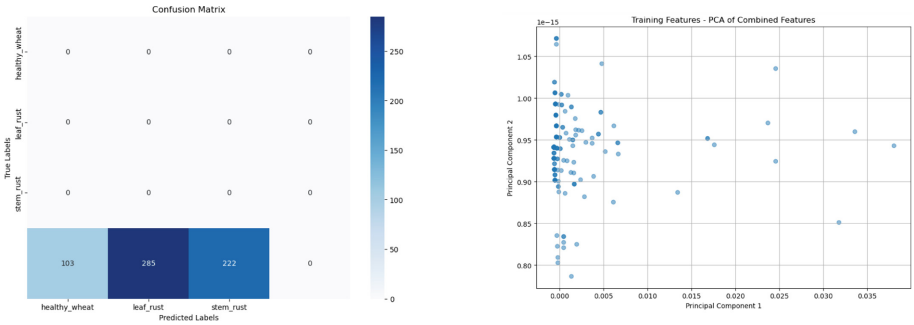
For instance, [7] used UAV-based classification to detect Cercospora leaf spot in sugar beets, achieving assuring accuracy but with limitations in adaptability across varying environmental conditions due to dependency on RGB color features only, which are sensitive to lighting variations. In another example, [14] developed a tea leaf disease model incorporating transfer learning, enhancing precision and recall; however, the model's specialization for a single disease type restricts its versatility to other diseases. Our approach, in contrast, utilizes a combination of HSV and GLCM texture features, enabling varied application across disease types while maintaining high performance with reduced computational demands.

Moreover, feature-based methods like those of [4], which evaluate multiple classification models for plant disease detection, struggle with balancing accuracy and computational efficiency. Our approach, using RF, shows comparable accuracy to CNN-based methods demanding fewer resources, which is crucial for deployment on mobile platforms. Finally, [9] applied deep learning for plant disease detection using a CNN, achieving high accuracy but facing limitations in real-time deployment due to the model's computational intensity. Our model addresses this limitation by achieving similar accuracy while being optimized for efficiency, making it more suitable for real-time applications.

7 Analysis



(a) Confusion matrix

(b) Training with PCA based features

Fig. 8. Comparison of confusion matrix and PCA based features.

The confusion matrix shows that the model is very accurate at classifying images of wheat rust (see Fig. 8(a)). The model correctly classified 285 images of leaf rust, 222 images of stem rust, and 103 images of healthy wheat (see Fig. 7). The model did not misclassified any images of wheat rust, it is well-trained and is able to distinguish between the different types of wheat rust. The confusion matrix evaluates the performance of a classification model and provides information

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on errors and improvement. Figure 8(b) depicts a scatter plot of two principal components. It shows the relationship between the two principal components for the training data and demonstrates a positive correlation between the them. The data points are clustered together, indicating that they have similar values for both principal components. This could be helpful for tasks like determining the dimensionality of the data, or selecting the features that are most important for prediction.

8 Conclusion and Future Work

This study develops an advanced system for detecting wheat leaf diseases, particularly leaf and stem rust, using both aerial and ground-level image data. By integrating HSV color features and GLCM-based texture analysis, the system accurately distinguishes healthy plants from those affected by rust diseases. The utilization of RF, SVM, and KNN classifiers led to robust model training, with RF achieving the best performance metrics such as precision, accuracy, recall, and F1-score confirming the model's efficiency in real-world scenarios. The promising results suggest potential for further improvements through dataset expansion and exploration of additional machine learning techniques to enhance detection accuracy.

This research aims to create a user-friendly mobile app or web tool for real-time monitoring of rust disease in crops, adaptable for farmers in the field. This innovative system will classify rust infections into early, mid, and late stages, giving insights into the disease's severity and spread. Additionally, it will feature a visual guide that shows the symptoms of each rust stage, helping farmers recognize and understand the disease. By providing real-time data and practical insights, it will empower farmers to take proactive steps in managing rust disease, ultimately reducing crop losses and boosting agricultural productivity.

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Chapter 33

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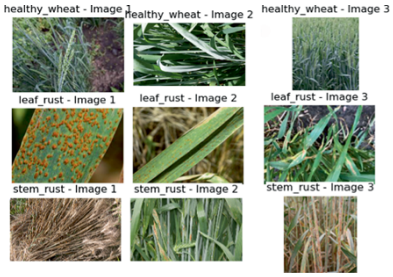
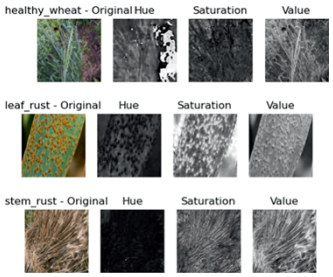
Page no	Fig/Photo	Thumbnail	Alt-text Description
5	Fig1		A grid of nine images showing different conditions of wheat plants. The top row displays three images labeled "healthy_wheat," showing lush, green wheat plants. The middle row shows three images labeled "leaf_rust," depicting wheat leaves with orange-brown spots indicating rust infection. The bottom row contains three images labeled "stem_rust," featuring wheat stems with rust-colored streaks and damage. Each image is labeled with its respective condition and number.
5	Fig2		A series of five images showing wheat plants. The first two images depict healthy wheat with green leaves. The third and fourth images show wheat leaves affected by leaf rust, characterized by orange-brown spots. The fifth image displays wheat stems with stem rust, featuring elongated brown lesions. Each image is labeled accordingly at the top.
7	Fig3		A series of images showing wheat conditions and their color properties. The top row displays "healthy wheat" with four images: the original photo, and its hue, saturation, and value representations. The middle row shows "leaf rust" with similar four images, highlighting rust spots on a leaf. The bottom row depicts "stem rust" with four images, focusing on affected wheat stems. Each set illustrates the original condition and its color analysis.

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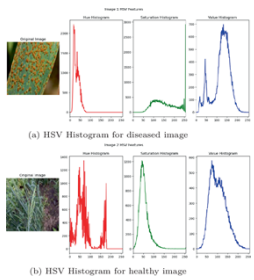
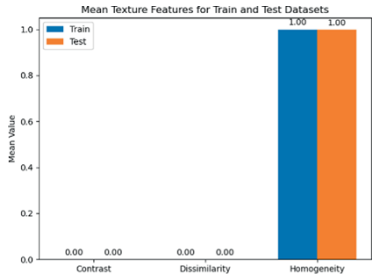
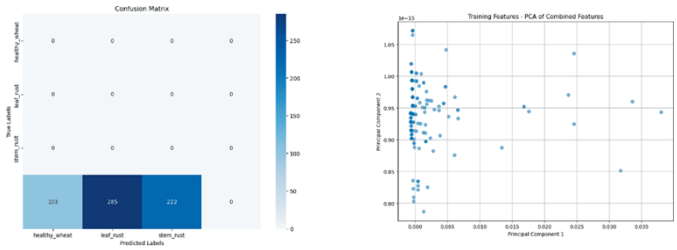
8	Fig4		The image consists of two sets of HSV histograms for plant images. The top row (a) shows histograms for a diseased plant image, with separate charts for hue, saturation, and value. The bottom row (b) displays histograms for a healthy plant image, also with separate charts for hue, saturation, and value. Each set includes a photograph of the plant on the left, followed by the three histograms. The histograms illustrate the distribution of color components in the images.
9	Fig5		Bar chart titled "Mean Texture Features for Train and Test Datasets" showing three categories: Contrast, Dissimilarity, and Homogeneity. Both the Train and Test datasets have a mean value of 1.00 for Homogeneity, while Contrast and Dissimilarity have mean values of 0.00. The Train dataset is represented in blue, and the Test dataset in orange.
10	Fig6		Flow chart illustrating a data processing workflow. It begins with "Input Data," followed by "Data Preprocessing," then "Feature Extract." The process continues to "Model Training," leading to "Prediction and Evaluation," and concludes with "Visualization." Arrows indicate the flow direction between each step. Icons accompany some steps, such as a database icon for "Input Data" and a gear icon for "Feature Extract."

Page no	Fig/Photo	Thumbnail	Alt-text Description
11	Fig7	(a) leaf rust (b) healthy (c) stem rust	Three images of wheat plants with predicted labels. (a) Leaf rust: Close-up of a leaf with orange-brown spots indicating rust infection. (b) Healthy: Green wheat plants with no visible signs of disease. (c) Stem rust: Wheat stems with brownish discoloration, suggesting rust infection. Each image includes a predicted label above and a description below.
12	Fig8	 (a) Confusion matrix (b) Training with PCA based features	The image consists of two panels. Panel (a) is a confusion matrix showing classification results with predicted labels on the x-axis and true labels on the y-axis, including categories like healthy wheat, leaf rust, and stem rust. The matrix highlights values such as 103, 285, and 222. Panel (b) is an X-Y chart displaying a scatter plot of training features using PCA (Principal Component Analysis) of combined features, with Principal Component 1 on the x-axis and Principal Component 2 on the y-axis. Data points are scattered across the chart, indicating the distribution of features.